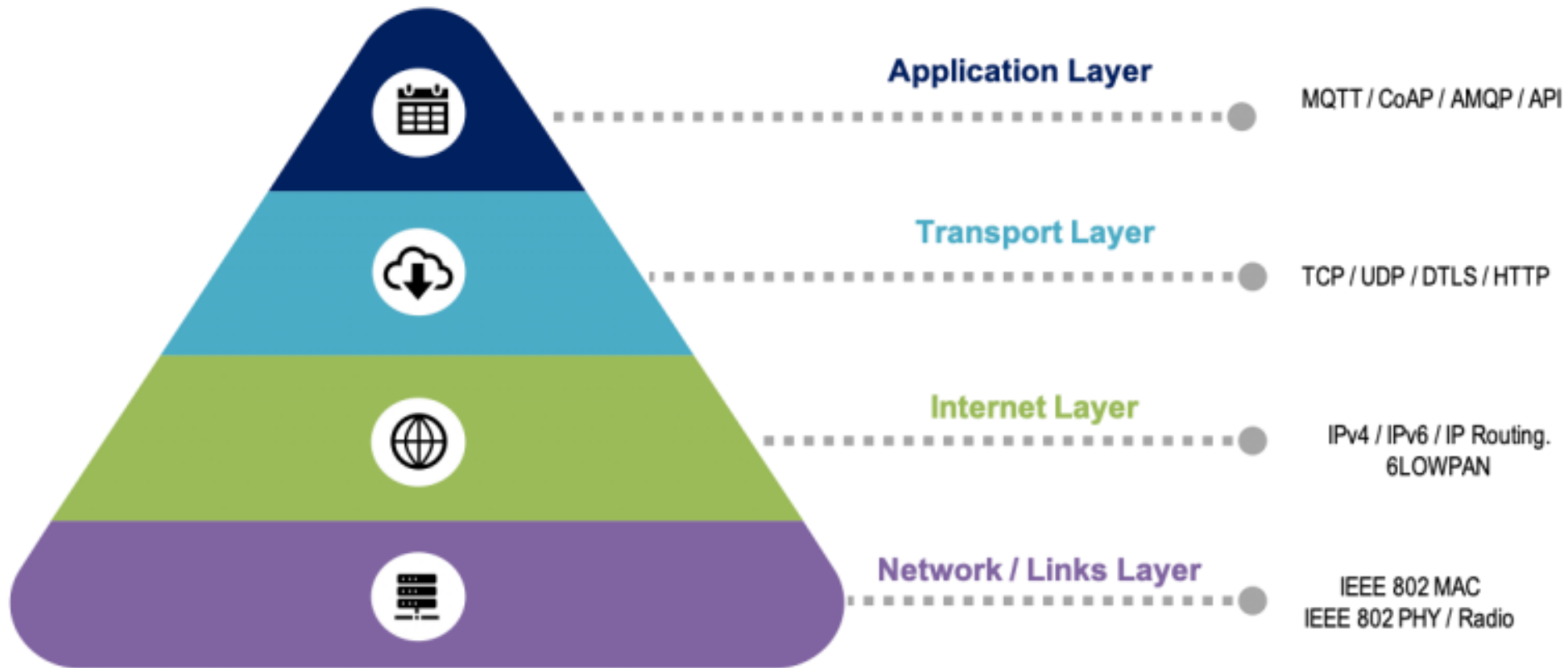


IoT Networks

What is IoT Network?

- **IoT Network:** Refers to communication technologies used by IoT devices to share data within reachable distances.
- Various Types of IoT Networks are available for IoT devices/sensors to communicate, crucial for real-time data collection and accessing insights.
- **Selection Strategy:** Choosing the proper networking protocol based on requirements is critical for IoT implementation.
- **IIoT (Industrial Internet of Things):** A sub-specification of IoT designed for industry automation and Smart manufacturing.
- **Network Choice:** Vital for IIoT success in Smart Manufacturing, emphasizing the importance of selecting appropriate wireless networks.

Layers of Network



Layers of Network

Network/Links Layer:

Function: This layer primarily deals with the physical connection between devices on the same network. It is responsible for the transmission and reception of data frames over the physical network medium.

Key Activities: Frame addressing, error detection, and media access control.

Examples: Ethernet, Wi-Fi, Bluetooth.

Internet Layer:

Function: The Internet Layer is responsible for logical addressing, routing, and forwarding of data packets between different networks. It enables communication between devices on different subnets.

Key Activities: IP address assignment, routing, packet forwarding.

Examples: Internet Protocol (IP).

Layers of Network

Transport Layer:

Function: This layer ensures end-to-end communication, providing reliable and efficient data transfer between devices. It manages flow control, error correction, and retransmission of lost or corrupted data.

Key Activities: Segmentation of data, error checking, flow control.

Examples: Transmission Control Protocol (TCP), User Datagram Protocol (UDP)

Application Layer:

Function: The Application Layer serves as the interface between the network and the software applications. It provides network services directly to end-users and supports communication services for various applications.

Key Activities: File transfers, email services, network browsing, and other high-level protocols.

Examples: Hypertext Transfer Protocol (HTTP), File Transfer Protocol (FTP), Simple Mail Transfer Protocol (SMTP).

Types of IoT Networks

Major wireless network types that can facilitate IoT applications and IoT sensor deployment in industries.



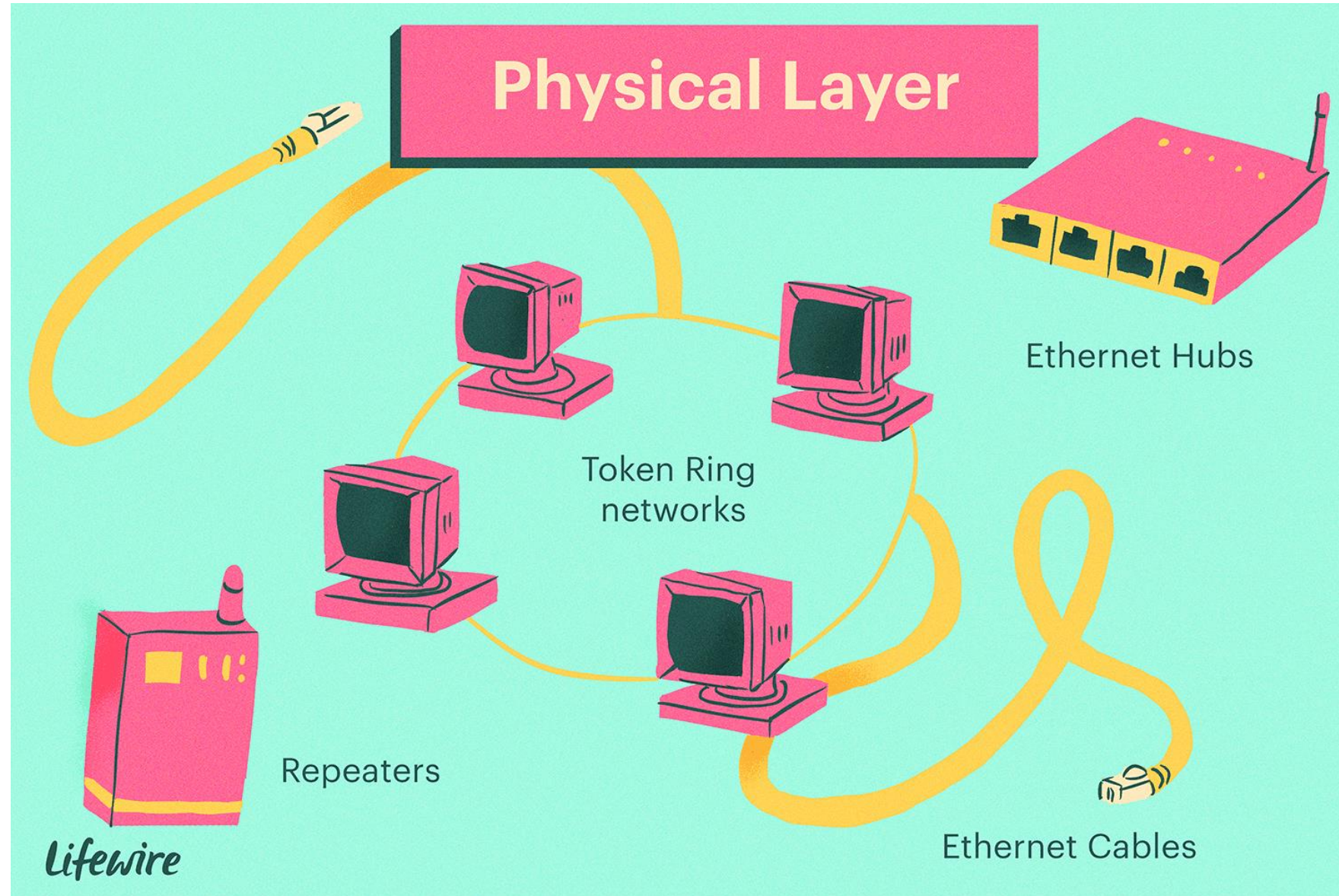
OSI Model

- The OSI (Open Systems Interconnection) model is a conceptual framework that standardizes the functions of a telecommunication or computing system into seven abstraction layers.
- Each layer is designed to perform specific tasks, and the model facilitates interoperability and communication between different systems.

OSI Model

7	Application Layer	Human-computer interaction layer, where applications can access the network services
6	Presentation Layer	Ensures that data is in a usable format and is where data encryption occurs
5	Session Layer	Maintains connections and is responsible for controlling ports and sessions
4	Transport Layer	Transmits data using transmission protocols including TCP and UDP
3	Network Layer	Decides which physical path the data will take
2	Data Link Layer	Defines the format of data on the network
1	Physical Layer	Transmits raw bit stream over the physical medium

Physical Layer of OSI Model:



Physical Layer of OSI Model:

- The Physical Layer is the lowest layer in the OSI model, responsible for the actual physical connection between devices.
- It deals with the transmission and reception of raw, unstructured bits over a physical medium

Function:

- Transmits individual bits from one node to another.
- Defines the characteristics of the hardware, such as cables, connectors, and the transmission medium.

Activities:

Bit Encoding: Involves converting digital data into a physical signal. Different encoding schemes (e.g., NRZ, Manchester encoding) are used.

Physical Topology: Defines how devices are connected physically (e.g., bus, star, ring topologies).

Transmission Rate: Specifies the speed at which bits are transmitted over the medium (e.g., Mbps, Gbps).

Media Types:

Guided Media: The physical medium is guided and includes cables like twisted pair, coaxial, and fiber-optic cables.

Unguided Media: The physical medium is unguided, and examples include wireless communication.

Devices:

Bit Synchronization:

Ensures that the sender and receiver have synchronized clocks to interpret the transmitted bits correctly.

Error Detection:

Basic error detection mechanisms are often included at this layer, but error correction is typically handled at higher layers.

Voltage Levels:

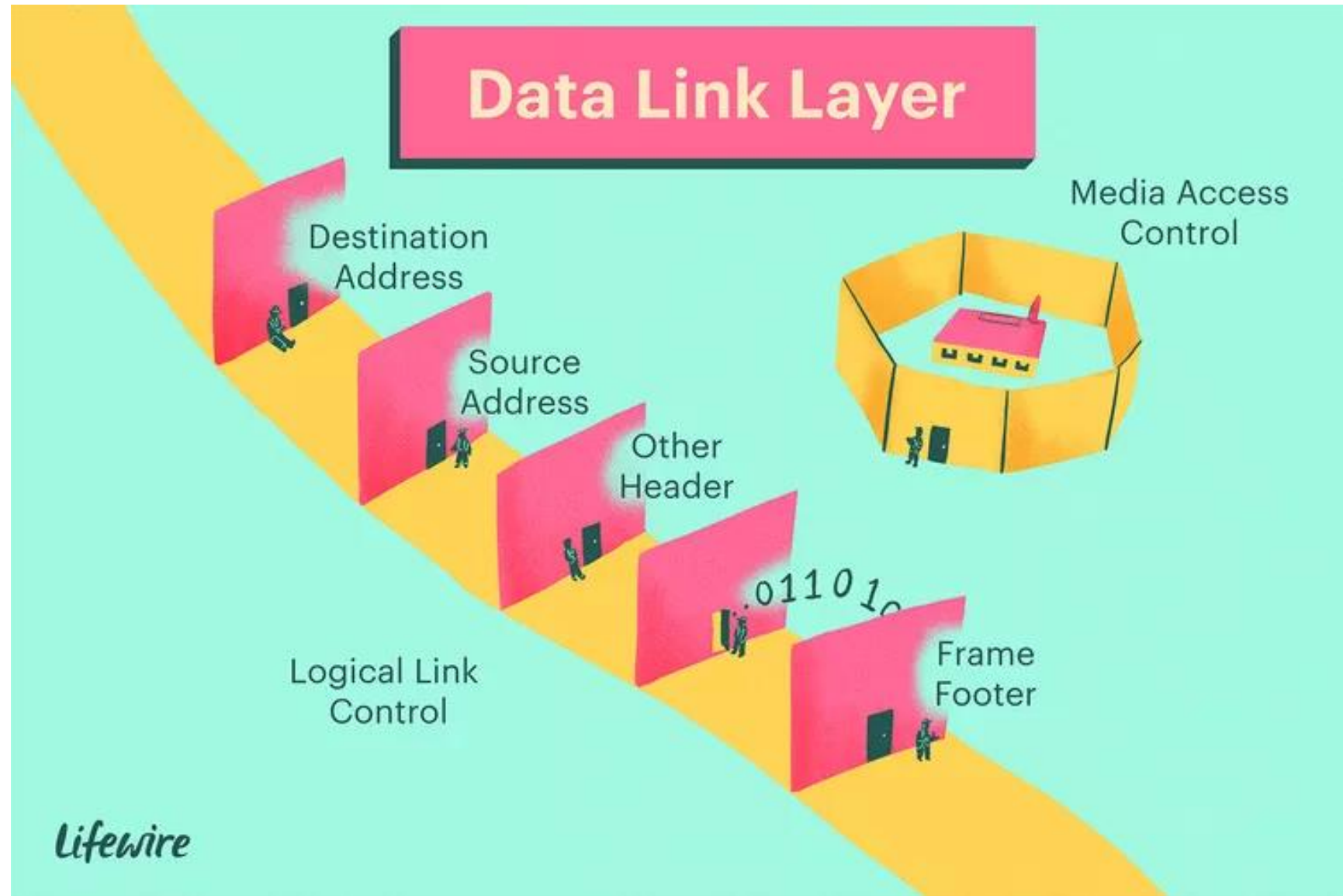
Defines the voltage levels representing binary 0 and 1, ensuring reliable signal interpretation.

Example Protocols:

Ethernet: Defines the physical and data link layer specifications for wired LANs.

RS-232: Defines the physical and electrical characteristics of the serial communication interface.

- When obtaining data from the Physical layer, the Data Link layer checks for physical transmission errors and packages bits into data frames.
- The Data Link layer also manages physical addressing schemes such as **MAC addresses** for Ethernet networks, controlling access of network devices to the physical medium.
- Because the Data Link layer is the most complex layer in the OSI model, it is often divided into two parts: the **Media Access Control** sub-layer and the **Logical Link Control** sub-layer.



Data Link Layer:

•Function:

- Facilitates error-free communication between directly connected nodes.
- Manages access to the physical medium for shared communication.

•Activities:

- Frames data for transmission and assigns unique addresses (MAC addresses).
- Implements error detection (e.g., CRC) and flow control mechanisms.

•MAC (Media Access Control):

- Defines rules for accessing the shared communication medium.
- Controls simultaneous transmission to prevent collisions.

•Logical Link Control (LLC):

- Establishes, maintains, and terminates links between devices.
- Manages flow control and error recovery at the logical link level.

•Frame Types:

- Data Frames: Carry user data.
- Control Frames: Manage flow and error control.
- Acknowledgment Frames: Confirm successful receipt.

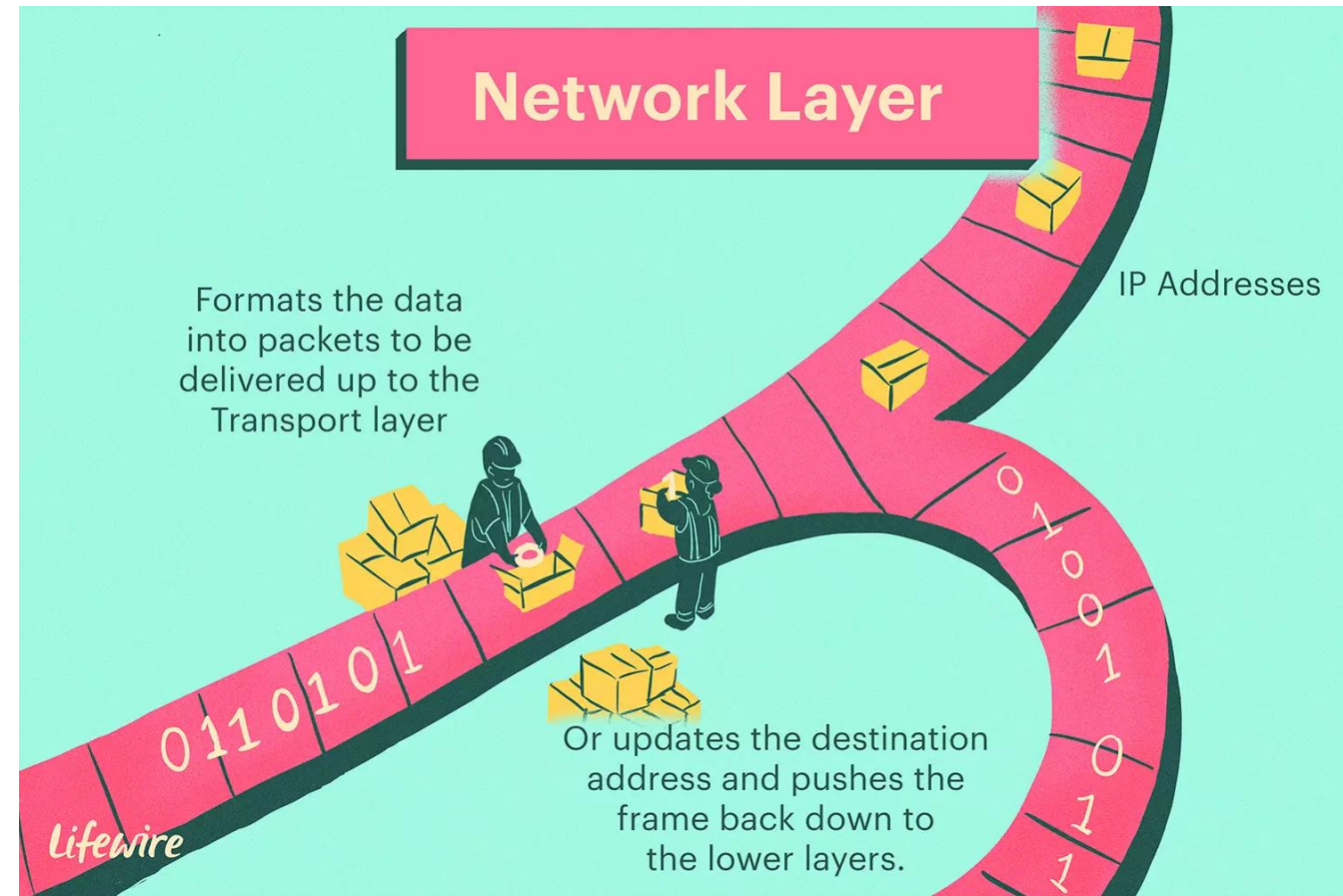
•Devices:

- Bridges, Switches: Filter and forward frames based on MAC addresses.
- NICs: Implement Data Link Layer protocols for device connectivity.

•Example Protocols:

- Ethernet: Common LAN protocol at Data Link and Physical Layers.
- PPP (Point-to-Point Protocol): Used for point-to-point communication.

- The Network layer adds the concept of routing above the Data Link layer.
- When data arrives at the Network layer, the source and destination addresses contained inside each frame are examined to determine if the data has reached its final destination.
- If the data has reached the final destination, layer 3 formats the data into packets delivered to the Transport layer.
- Otherwise, the Network layer updates the destination address and pushes the frame down to the lower layers.
- To support routing, the Network layer maintains logical addresses such as IP addresses for devices on the network.
- The Network layer also manages the mapping between these logical addresses and physical addresses. In **IPv4** networking, this mapping is accomplished through the **Address Resolution Protocol (ARP)**; **IPv6** uses **Neighbor Discovery Protocol (NDP)**.



Network Layer of OSI Model:

•Function:

- Enables logical addressing and routing of data packets between different networks.
- Facilitates communication between devices on different subnets.

•Activities:

- Assigns IP addresses for identification and routing purposes.
- Routes data packets based on logical addressing.

•Key Features:

- **IP Addressing:** Unique identifiers assigned to devices.
- **Routing:** Determines optimal paths for data packet transmission.

•Packet Structure:

- Headers include source and destination IP addresses, protocol version, and time-to-live (TTL).

•Protocols:

- **Internet Protocol (IPv4, IPv6):** Core protocols for logical addressing and routing.
- **ICMP (Internet Control Message Protocol):** Manages error messages and diagnostics.

- **Routing Algorithms:**

- **Static Routing:** Pre-defined paths.
- **Dynamic Routing:** Adapts based on network conditions.

- **Devices:**

- **Routers:** Operate at the Network Layer to forward data between networks.
- **Layer 3 Switches:** Combine features of routers and switches.

- **Subnetting:**

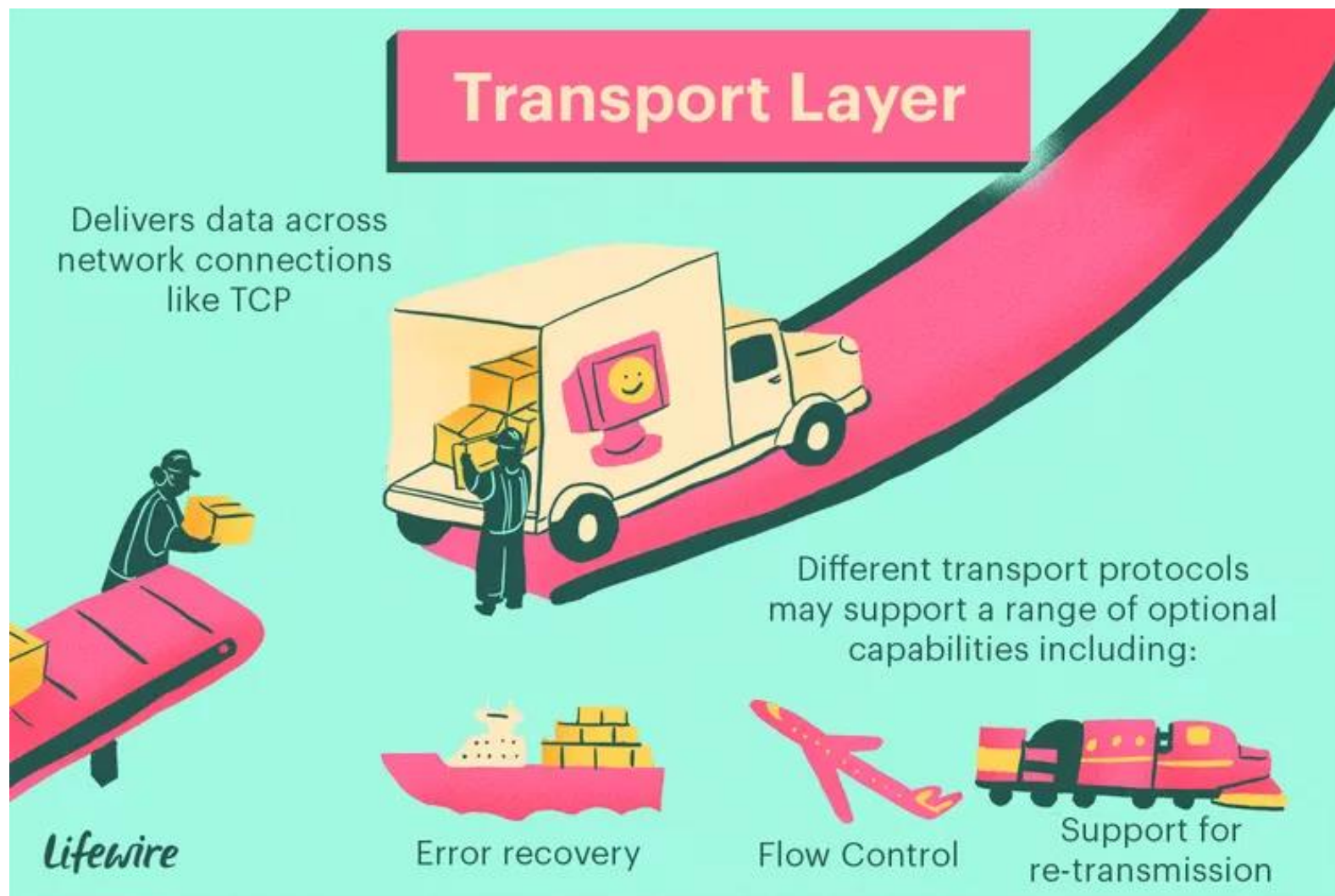
- Divides IP networks into smaller, manageable sub-networks.

- **Fragmentation and Reassembly:**

- Breaks large packets into smaller fragments for transmission and reassembles them at the destination.

- **Example Scenario:**

- A router receives a packet, examines the destination IP address, and forwards it to the appropriate network.



- The Transport Layer delivers data across network connections. TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) are the most common examples of Transport Layer 4 network protocols.
- Different transport protocols may support a range of optional capabilities, including error recovery, flow control, and support for re-transmission.

Transport Layer of OSI Model:

•Function:

- Ensures end-to-end communication and data transfer reliability.
- Manages flow control, error correction, and retransmission.

•Protocols:

- **TCP (Transmission Control Protocol):** Reliable, connection-oriented. Establishes a connection through a three-way handshake, guarantees reliable and ordered data delivery.
- **UDP (User Datagram Protocol):** Unreliable, connectionless. Suitable for real-time applications where speed is crucial, but may result in data loss.

•Key Features:

- **Segmentation:** Divides data into segments for transmission.
- **Error Checking:** Detects and corrects errors in data.

•Flow Control:

- Manages data transmission rate to prevent congestion.

•Connection Establishment and Termination:

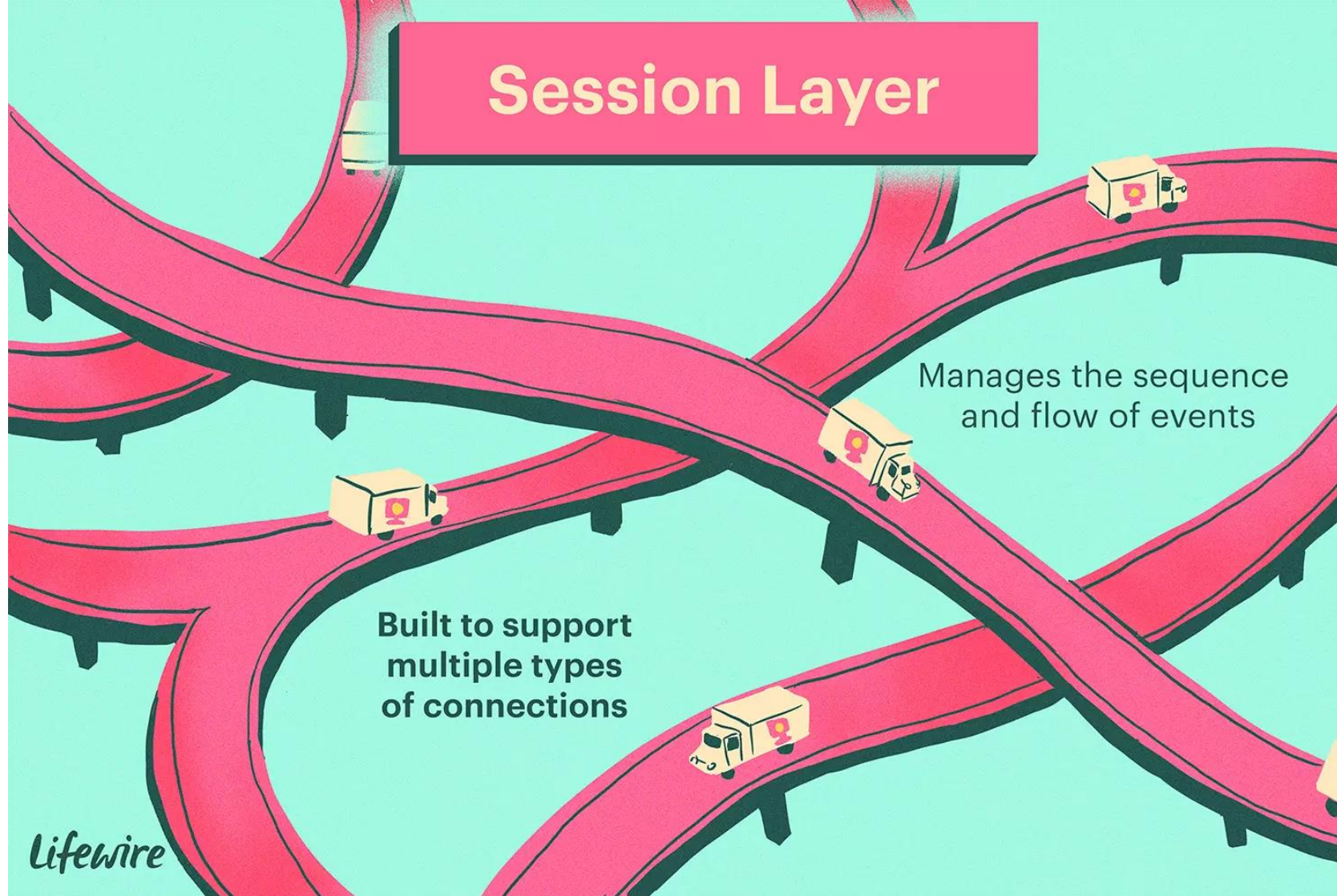
- TCP ensures a reliable connection with a three-way handshake.
- TCP guarantees a complete file transfer by acknowledging received segments and retransmitting if necessary.

•Port Numbers:

- Identify specific applications on devices.
- TCP and UDP use port numbers for addressing.

•Reliability:

- TCP guarantees reliable and ordered data delivery.
- UDP provides fast, connectionless transmission but may result in data loss.



- The Session Layer manages the sequence and flow of events that initiate and tear down network connections.
- At layer 5, it is built to support multiple types of connections that can be created dynamically and run over individual networks.

Presentation Layer of OSI Model:

•Function:

- Translates data between the Application Layer and the lower layers for seamless communication.
- Manages data format translation, encryption, and compression.

•Data Translation:

- Converts between different data formats, ensuring compatibility.

•Encryption and Compression:

- Encrypts sensitive data for secure transmission.
- Compresses data to reduce transmission overhead.

•Character Set Conversion:

- Converts between different character sets to support multilingual communication.

•Example Protocols:

- **JPEG (Joint Photographic Experts Group):** Manages image compression.
- **SSL/TLS (Secure Sockets Layer/Transport Layer Security):** Encrypts data for secure communication.

•Data Syntax:

- Defines the syntax and semantics of the information exchanged.

•Syntax Examples:

- **Text:** ASCII, Unicode.
- **Images:** JPEG, GIF.

•Error Detection:

- Detects and corrects syntax errors in transmitted data.

•Example Scenario:

- In a web browser, the Presentation Layer translates HTML data received from a server into a readable format for the user.

Application Layer



The Application layer supplies network services to end-user applications. Network services are protocols that work with the user's data. For example, in a web browser application, the Application layer protocol [HTTP](#) packages the data needed to send and receive web page content. This layer 7 provides data to (and obtains data from) the Presentation layer.

Application Layer of OSI Model:

•Function:

- Provides network services directly to end-users and supports communication services for various applications.
- Serves as the interface between the network and software applications.

•Network Services:

- Enables user access to the network (e.g., email, file transfer, network browsing).

•Protocols:

- **HTTP (Hypertext Transfer Protocol):** Facilitates web communication.
- **FTP (File Transfer Protocol):** Manages file transfers between devices.
- **SMTP (Simple Mail Transfer Protocol):** Handles email transmission.

•User Interface:

- Interacts with end-user applications, presenting data and receiving user input.

•Example Applications:

- Web browsers, email clients, file transfer applications.

•Data Exchange:

- Manages the transfer of data between different applications.

•Multiplexing and Demultiplexing:

- Combines multiple application-layer messages into a single communication stream.
- Separates incoming data back into individual messages.

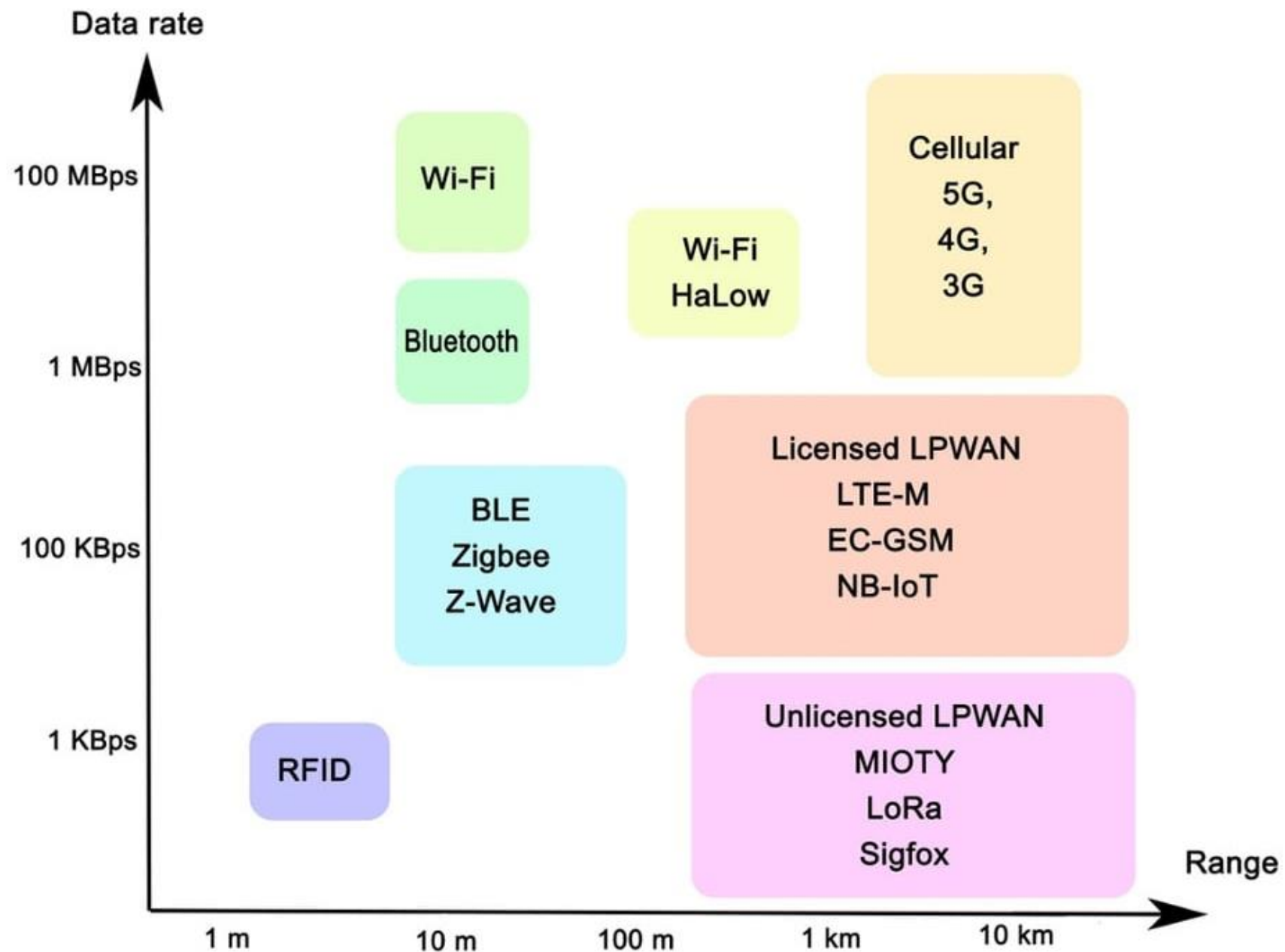
•Example Scenario:

- A user interacts with a web browser (application layer) to access a website, and HTTP facilitates communication between the browser and the server

IoT Communication Protocols

5 things to know about IoT protocols

-  **1** There is no one-size-fits-all IoT protocol
-  **2** Protocols purpose-made for IoT are seeing increased adoption
-  **3** Software is becoming more important to managing IoT connectivity
-  **4** “Ease of use” and “reliability” are most important when choosing a new protocol
-  **5** Decision making for IoT protocols has become a multi-stakeholder exercise

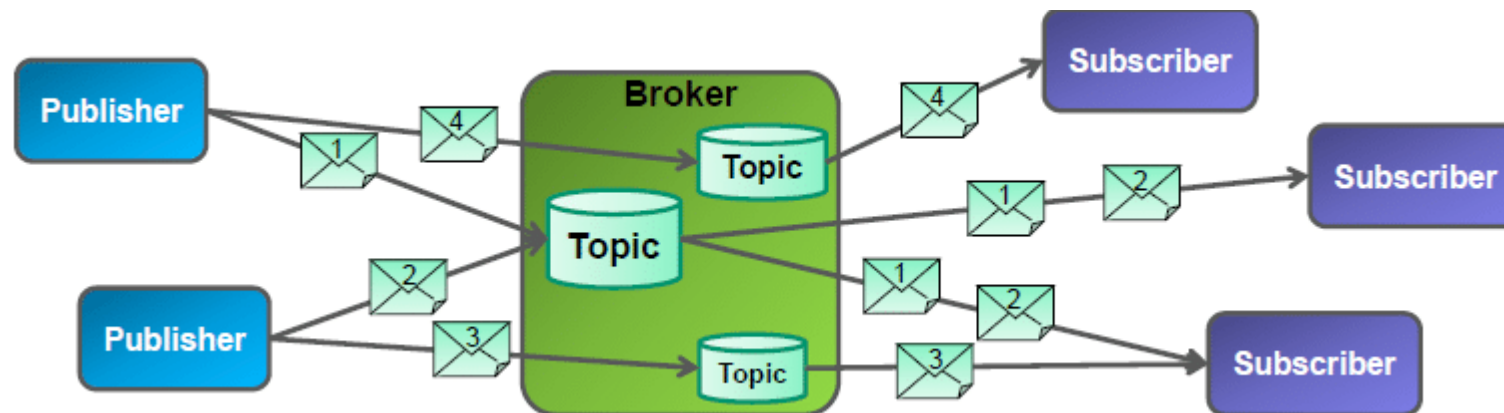




The application layer of a network enables network entities to identify and interact with each other.

MQTT (Message Queueing Telemetry Transport)

- The Enabler Of Smooth And Hassle-Free Information Exchange For An IoT Ecosystem
- MQTT is a lightweight communication protocol specifically designed for IoT and M2M applications.
- It's ideal for remote environments or applications with limited bandwidth
- MQTT uses a connection oriented publish/subscribe architecture, where MQTT applications can either publish (transmit) or subscribe to (receive) topics, and an MQTT broker passes information from the publishing client to the subscribed client.





The application layer of a network enables network entities to identify and interact with each other.

MQTT (Message Queueing Telemetry Transport)

Publish/Subscribe Model

- MQTT operates on a publish/subscribe paradigm.
- Publishers: Devices or applications send messages to specific "topics."
- Subscribers: Devices or applications subscribe to specific topics to receive relevant messages.

Broker

- The MQTT broker serves as a central hub.
- It receives messages from publishers and forwards them to subscribers based on their topic subscriptions.
- Facilitates communication between devices without direct connections.

Quality of Service (QoS) Levels

MQTT supports different QoS levels:

- 0: At most once delivery (no acknowledgment).
- 1: At least once delivery (guaranteed delivery but may be duplicated).
- 2: Exactly once delivery (ensures no duplication).



The application layer of a network enables network entities to identify and interact with each other.

MQTT (Message Queueing Telemetry Transport)

Retained Messages

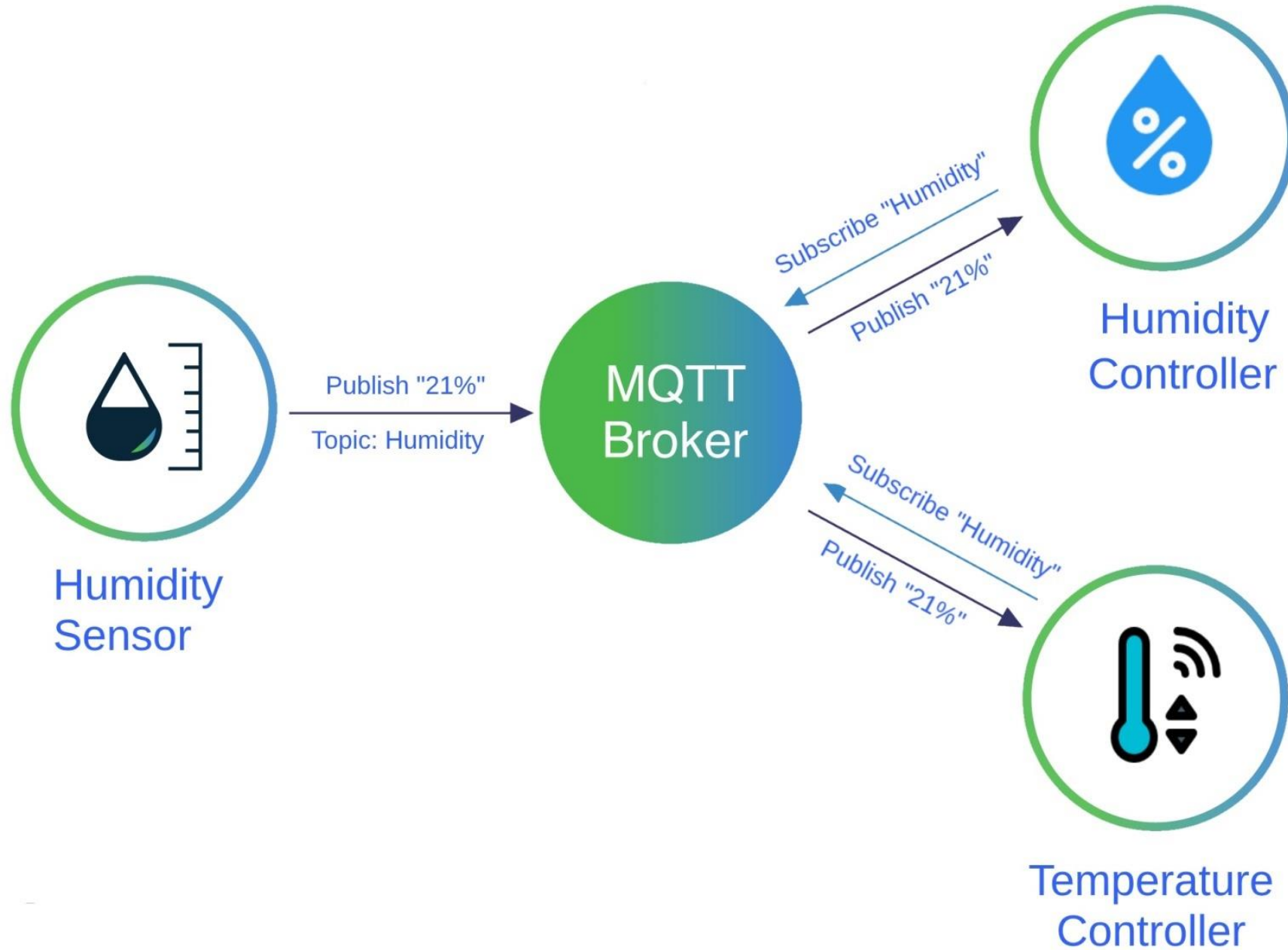
- A unique feature where the broker stores the last message on a topic.
- New subscribers immediately receive the latest message when they subscribe to a topic.

1. Lightweight and Efficiency

1. Designed for minimal bandwidth and resource usage, making it suitable for IoT devices with limited capabilities.

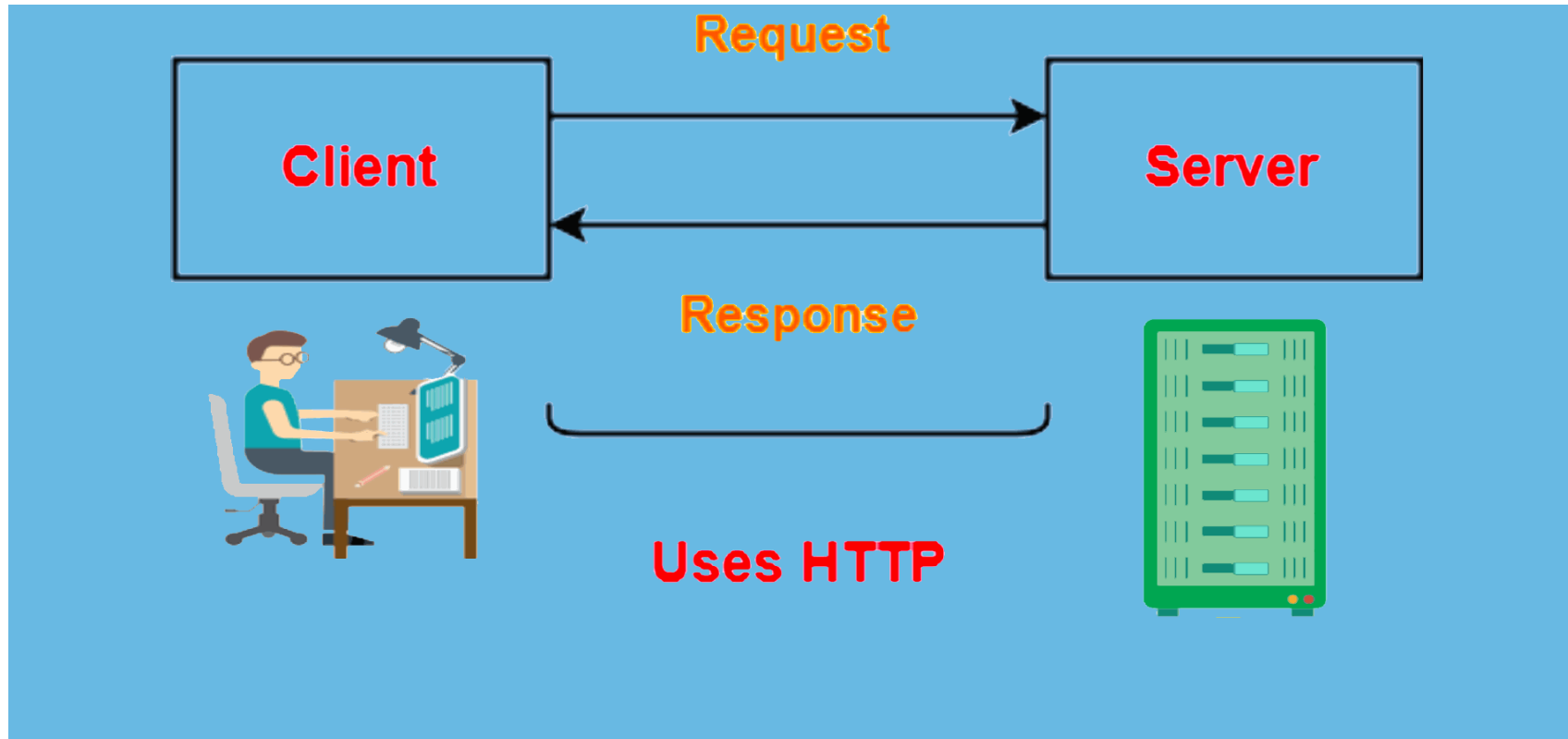
2. Use Cases

1. Widely adopted in IoT scenarios, home automation, and industrial applications.
2. Particularly beneficial in situations where low bandwidth and efficient communication are crucial.



HTTP (HyperText Transfer Protocol)

- HTTP (HyperText Transfer Protocol) is the foundation of data communication on the World Wide Web.
- It's an application layer protocol that governs the transfer of hypertext, which includes text, images, videos, and other multimedia content.



HTTP (HyperText Transfer Protocol)



Client-Server Communication

Client: Initiates requests to a server.

Server: Listens for requests, processes them, and sends back responses.

Stateless Protocol

Each request from a client to a server is independent, and the server does not retain any information about the client's state between requests.

Enables scalability and simplicity in web communication.

Request-Response Model

Request: Sent by the client to the server, specifying the desired action (e.g., GET for retrieving data).

Response: Sent by the server to the client, containing the requested information along with status information.

HTTP (HyperText Transfer Protocol)

Methods

GET: Retrieve data from the server.

POST: Send data to be processed to a specified resource.

PUT: Update a resource on the server.

DELETE: Remove a resource on the server.

And more, each serving a specific purpose.

Status Codes

Three-digit codes in response headers indicating the status of the request (e.g., 200 OK, 404 Not Found, 500 Internal Server Error).

While HTTP is not inherently secure, HTTPS (HTTP Secure) encrypts the data transmission, enhancing security.

Use Cases

Fundamental for web browsing, data retrieval, and interactions on the internet.



WebSocket



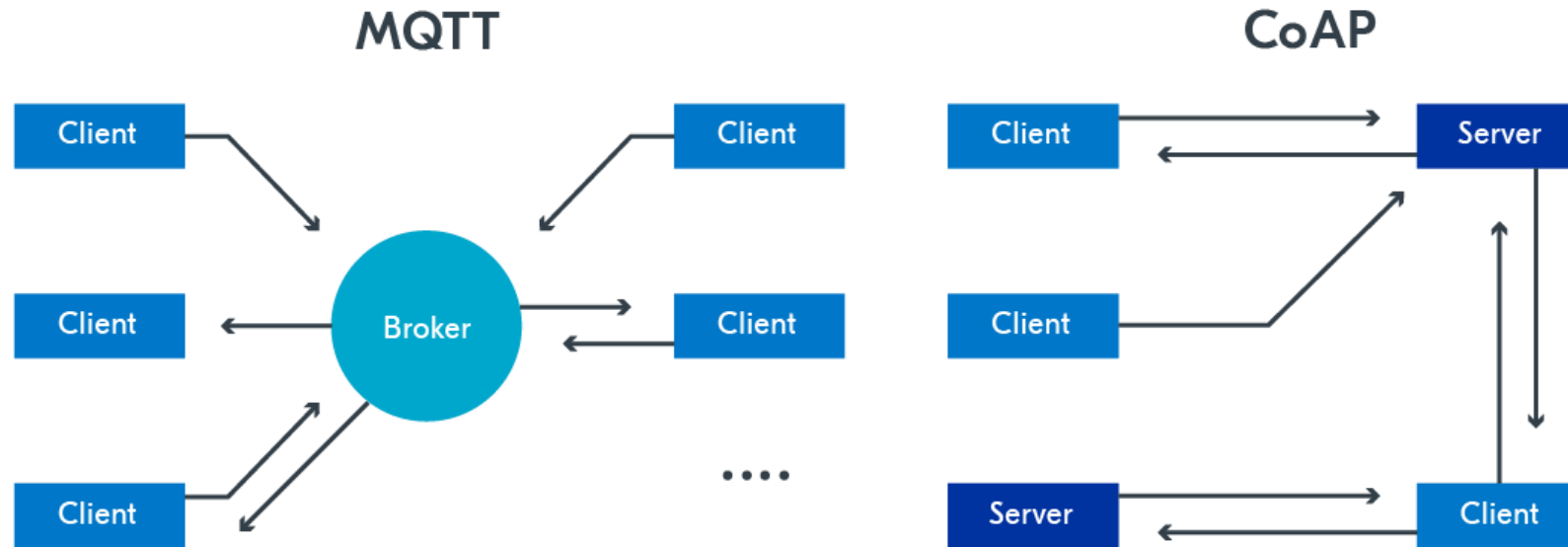
WebSocket

- **WebSocket** is a bi-directional communication protocol designed to quickly send large quantities of data in web applications.
- A **WebSocket** establishes a connection between client and server and therefore after the initial connection establishment—every single message only has small overhead.
- Devices and servers can simultaneously transmit and receive data in real-time, making this protocol best-suited for IoT applications where low latency is critical, communication happens frequently, and data consumption is less important.



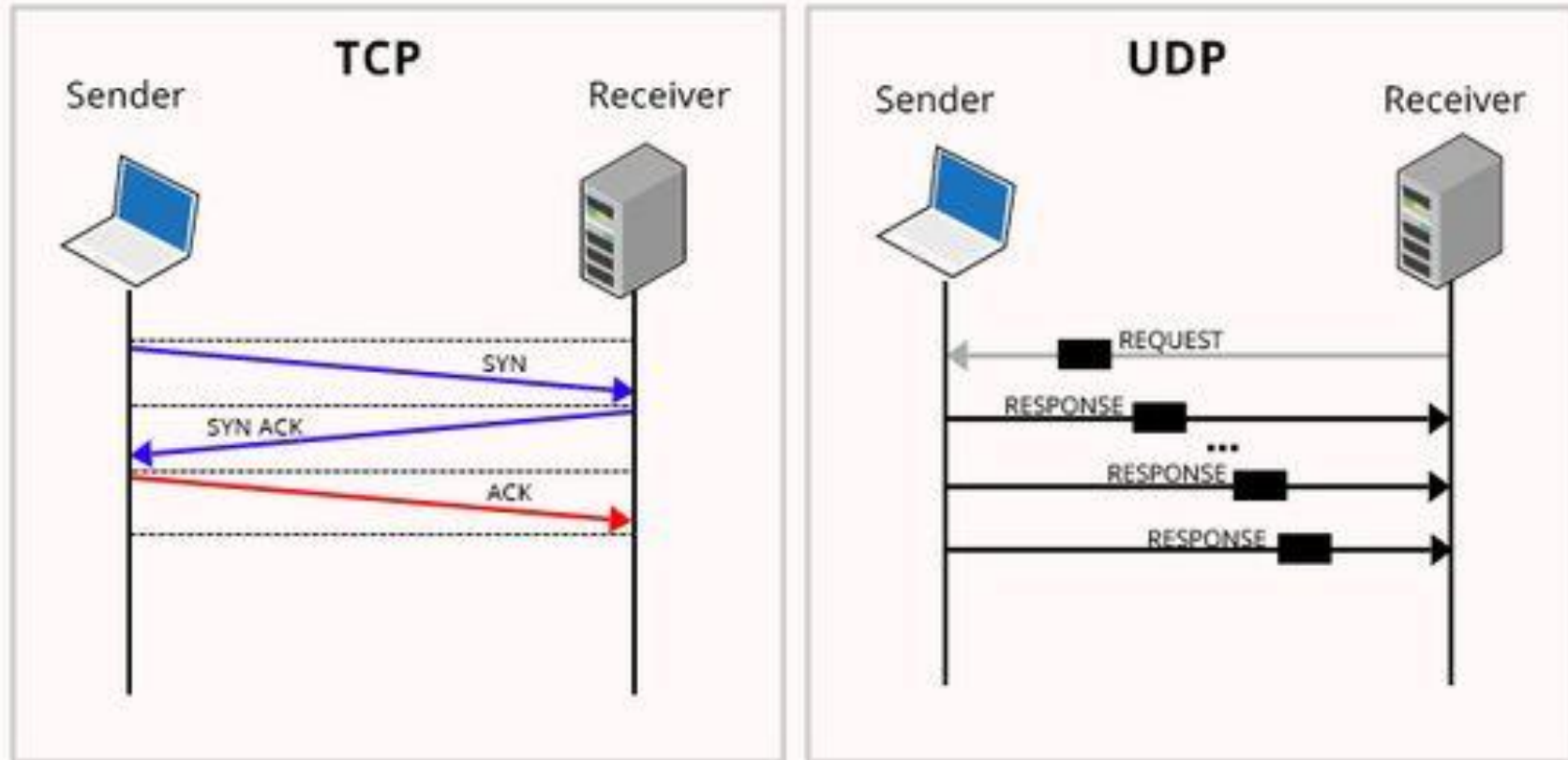
Constrained Application Protocol (CoAP)

- Constrained Application Protocol (CoAP) is designed for low-power, lossy networks, also known as “constrained” networks.
- CoAP is usually paired with User Datagram Protocol (UDP) which makes it highly efficient, making it appealing for IoT applications where battery conservation is important.
- For example, it’s often used in smart meter communications. CoAP can also use TCP or SMS as transport mechanism.



TCP (Transmission Control Protocol) and UDP (User Datagram Protocol)

TCP Vs UDP Communication



TCP (Transmission Control Protocol) and UDP (User Datagram Protocol)

- **TCP (Transmission Control Protocol)**
- Transmission Control Protocol (TCP) sets the parameters for data exchanges between software applications, confirms what is being sent, where it's coming from, where it's going, and whether or not it arrived correctly.
- TCP prioritizes accuracy over speed, ensuring that data arrives in order, with minimal errors, and without duplication. If data gets lost in transmission, TCP requests the data packets be resent.

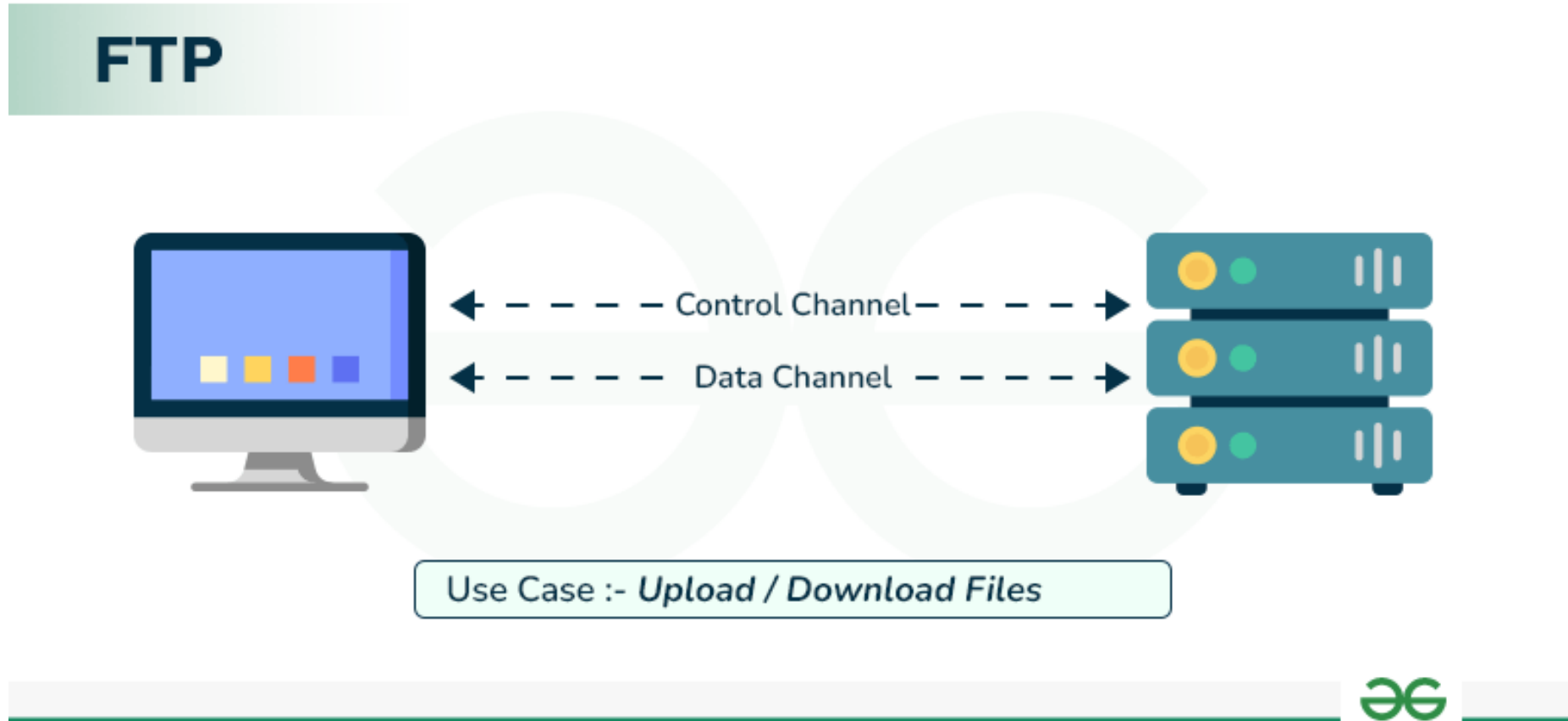
UDP (User Datagram Protocol)

- User Datagram Protocol (UDP) is a transport protocol that prioritizes speed over reliability, using a connectionless process to send data packets to a destination.
- Due to its low latency, UDP is ideal for time-sensitive use cases like video streaming, Voice over Internet Protocol (VoIP), video gaming, and Domain Name System (DNS) lookups.

TCP (Transmission Control Protocol) and UDP (User Datagram Protocol)

Feature	TCP	UDP
Connection Type	Connection-Oriented	Connectionless
Reliability	Reliable with error detection and correction	Unreliable, no error recovery mechanisms
Order of Delivery	Ensures correct order of delivery	Does not guarantee order of delivery
Flow Control	Implements flow control	No inherent flow control
Header Size	Larger header size due to additional features	Smaller header size, lower overhead
Use Cases	Suitable for applications prioritizing reliability	Ideal for real-time applications with low latency
Statefulness	Stateful, maintains connection state	Stateless, each datagram is independent

FTP (File Transfer Protocol)



FTP (File Transfer Protocol)

- **FTP (File Transfer Protocol)** is a standard network protocol used for the transfer of files from one host to another over a TCP-based network, such as the Internet.
- FTP works by opening two connections that link the computers trying to communicate with each other.
- One connection is designated for the commands and replies that get sent between the two clients, and the other channel handles the transfer of data.
- During an FTP transmission, there are four commands used by the computers, servers, or proxy servers that are communicating. These are “send,” “get,” “change directory,” and “transfer.”

What is FTP Useful For?

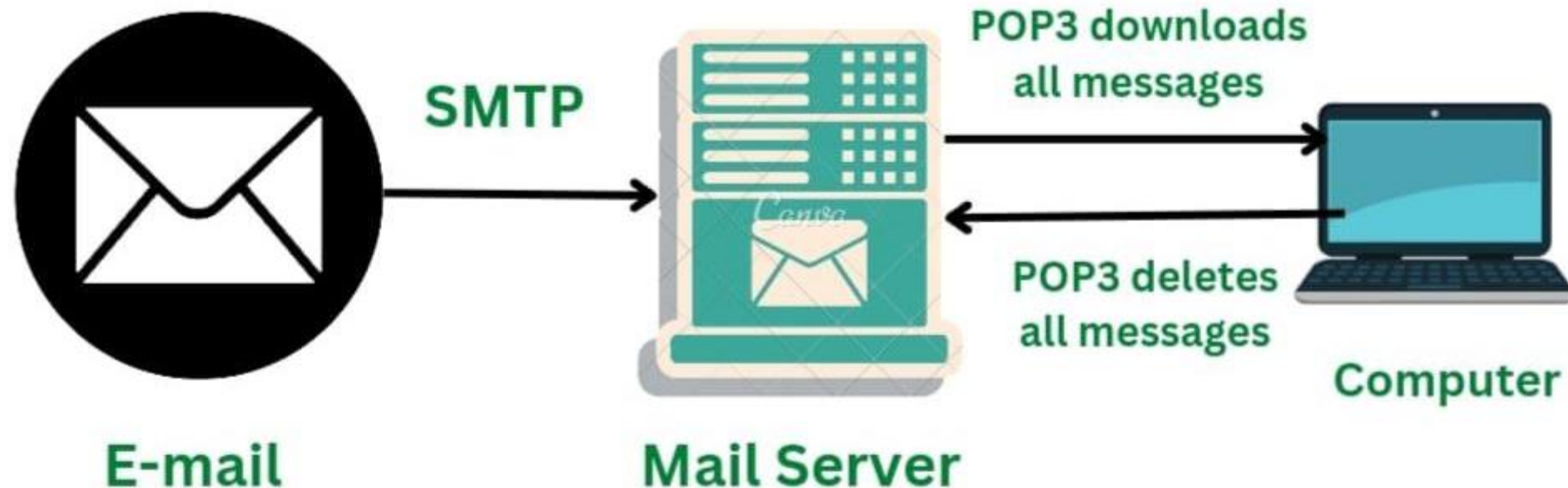
- One of the main reasons why modern businesses and individuals need FTP is its ability to perform large file size transfers. When sending a relatively small file, like a Word document, most methods will do, but with [FTP](#), you can send hundreds of gigabytes at once and still get a smooth transmission.

FTP (File Transfer Protocol)

Feature	FTP	HTTP
Purpose	Designed for file transfer between a client and a server	Primarily used for web browsing, serving web pages, and transferring hypermedia content
Transfer Type	Allows bidirectional transfer of files	Primarily supports unidirectional transfer of files from a server to a client
Connection Type	Connection-oriented, establishes a control connection for commands and a separate data connection for file transfers	Connectionless, creates a new connection for each request and response
Authentication	Supports various authentication mechanisms, including username and password	Authentication is typically handled through web forms or other mechanisms provided by the web application
Data Transfer Mode	Supports two modes: active and passive	Uses a single connection for data transfer, simplifying the communication process
Security	FTP lacks inherent security features, but FTPS (FTP Secure) and SFTP (SSH File Transfer Protocol) provide secure alternatives	HTTP lacks inherent security, but HTTPS (HTTP Secure) encrypts data transmission, ensuring confidentiality and integrity
Use Cases	Ideal for transferring large files, managing remote files, and collaborating on file-based projects	Primarily used for accessing and retrieving web content, including text, images, and multimedia

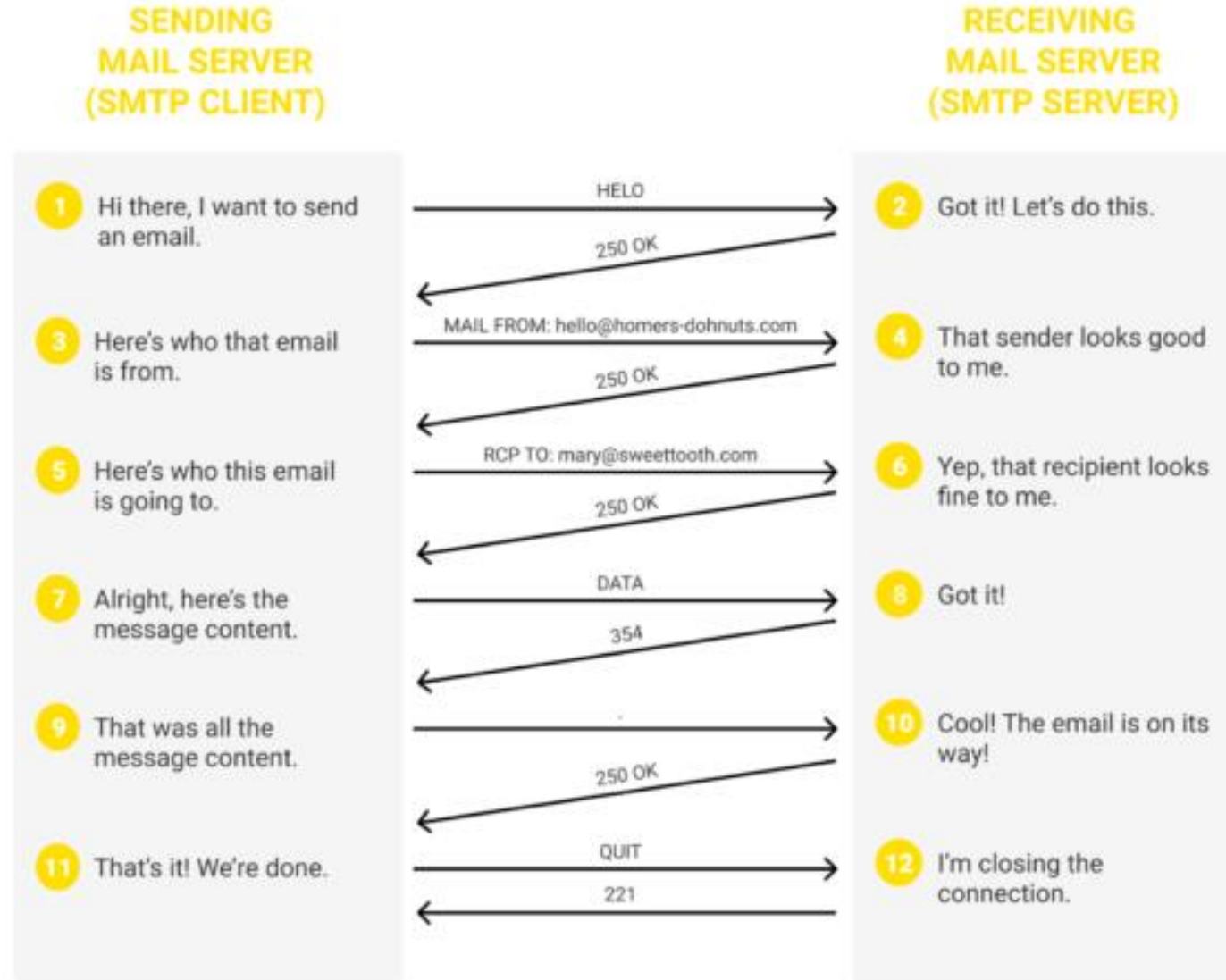
POP3 (Post Office Protocol 3)

- Retrieves emails from a server to a client device.
- Primarily used for downloading emails.
- Operates on the application layer of the OSI model.
- Uses port 110 for communication (unencrypted) and port 995 (encrypted).
- Typically deletes emails from the server after retrieval.



SMTP (Simple Mail Transfer Protocol)

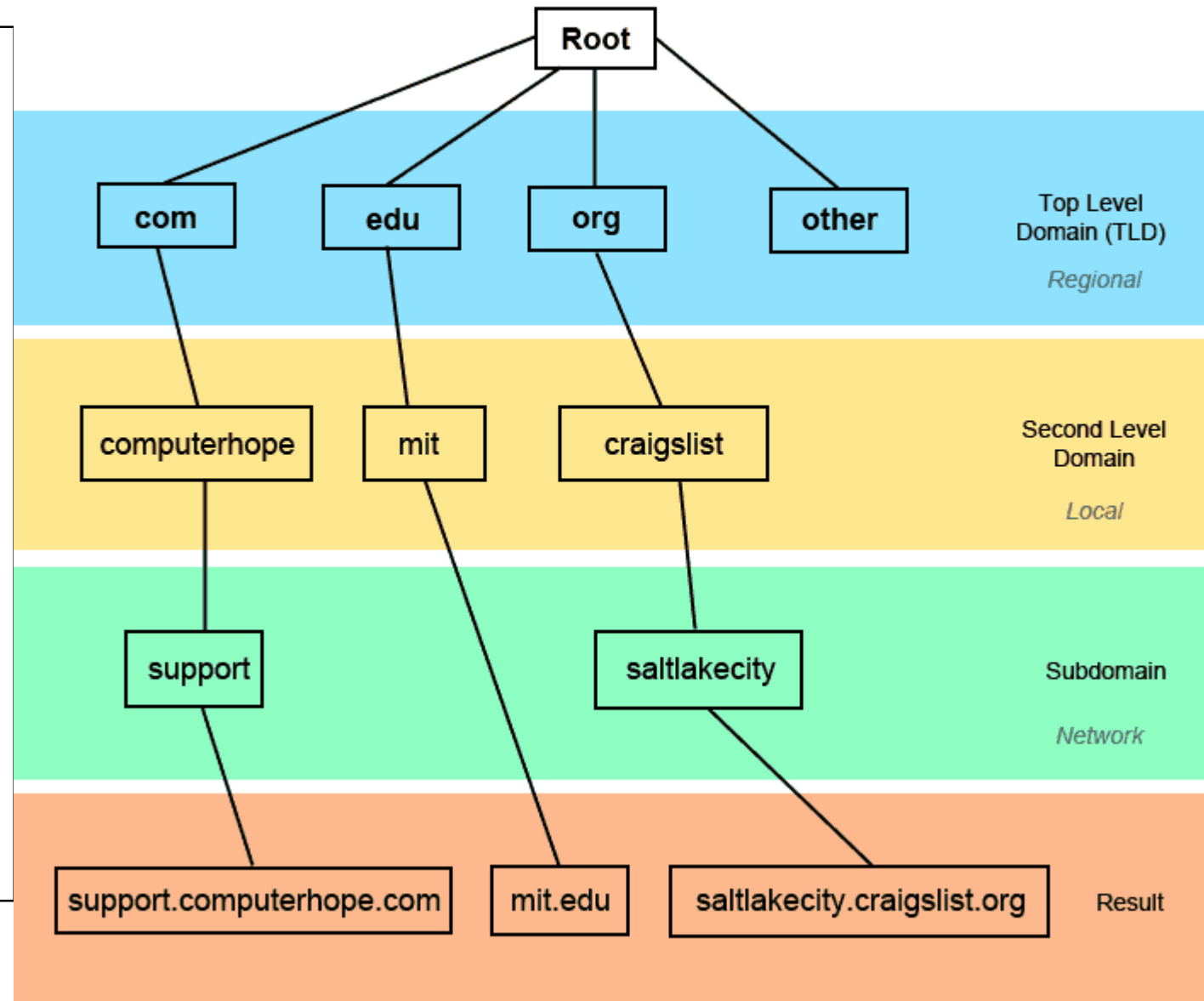
- Facilitates email transmission between servers.
- Standard protocol for sending emails.
- Operates on the application layer of the OSI model.
- Utilizes port 25 for communication.
- Essential for the functioning of email services.



DNS (Domain Name System)

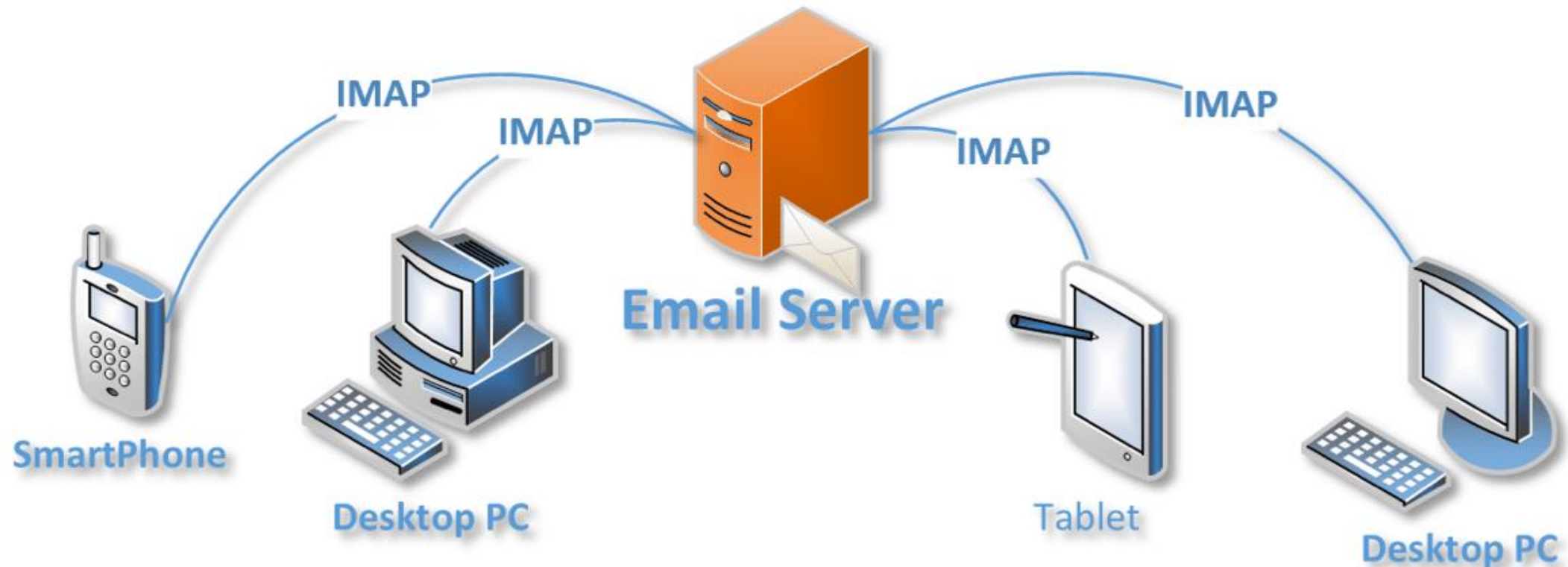
- Translates human-readable domain names into IP addresses.
- Fundamental for web browsing and internet navigation.
- Operates on the application layer of the OSI model.
- Utilizes both UDP (port 53) and TCP (port 53) for communication.
- Hierarchical structure with authoritative name servers.

Domain Naming Hierarchy



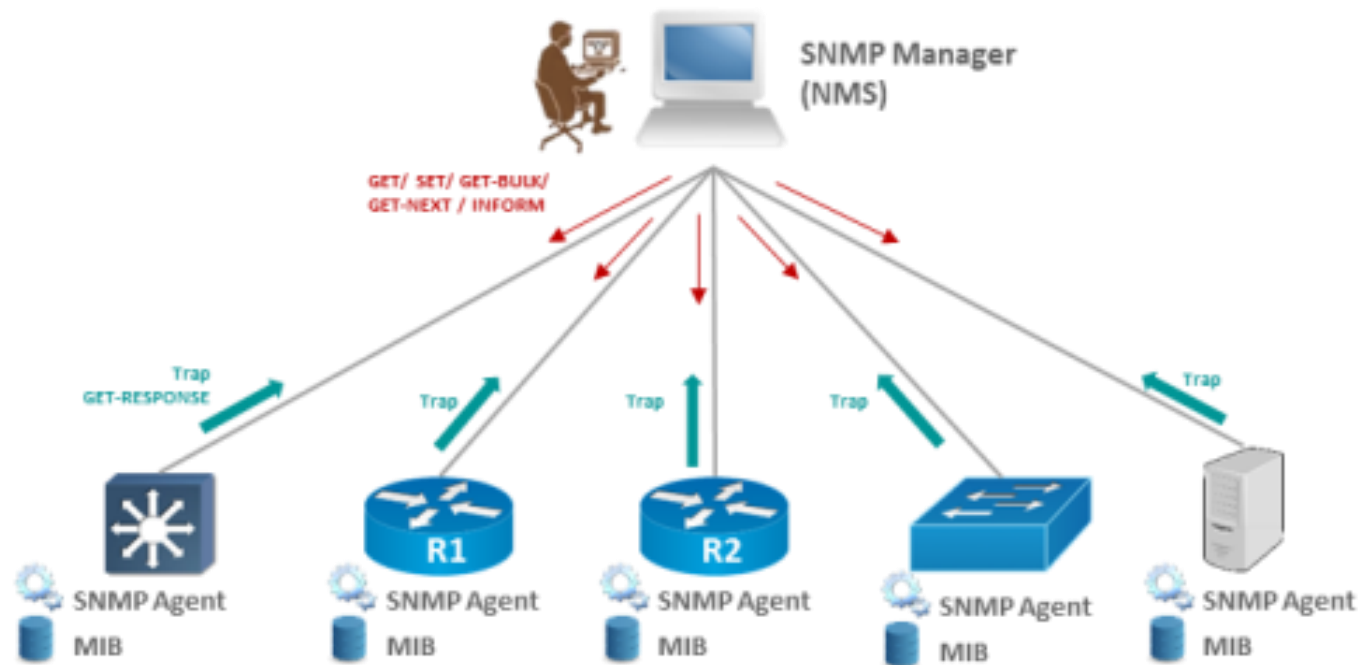
IMAP (Internet Message Access Protocol)

- Allows emails to be stored on a mail server.
- Enables synchronization of emails across multiple devices.
- Operates on the application layer of the OSI model.
- Uses port 143 for communication (unencrypted) and port 993 (encrypted).
- Retains emails on the server, offering flexibility in email management.



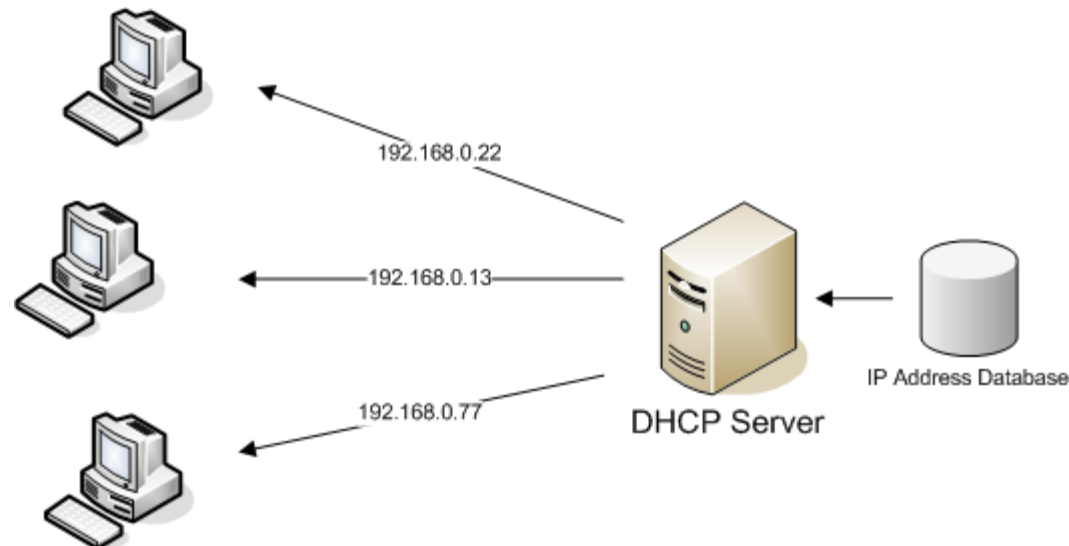
SNMP (Simple Network Management Protocol)

- Manages and monitors network devices.
- Collects and organizes information for network administration.
- Operates on the application layer of the OSI model.
- Utilizes UDP as the transport protocol.
- Enables the centralization of network monitoring and control.



DHCP (Dynamic Host Configuration Protocol)

- Automatically assigns IP addresses to devices on a network.
- Simplifies network configuration and management.
- Operates on the application layer of the OSI model.
- Uses UDP (port 67 for server, port 68 for client) for communication.
- Enables dynamic allocation of IP addresses and network parameters.



ICMP (Internet Control Message Protocol)

- Used for network diagnostics and error reporting.
- Includes echo requests and replies (ping) for connectivity testing.
- Operates on the network layer of the OSI model.
- Part of the Internet Protocol suite (IP).
- ICMP messages inform routers and hosts of network conditions.

